

Practical No : 3

Tutorial Title: Write a program in C++ to sort the numbers in an array using separate functions for read, display, sort and swap.

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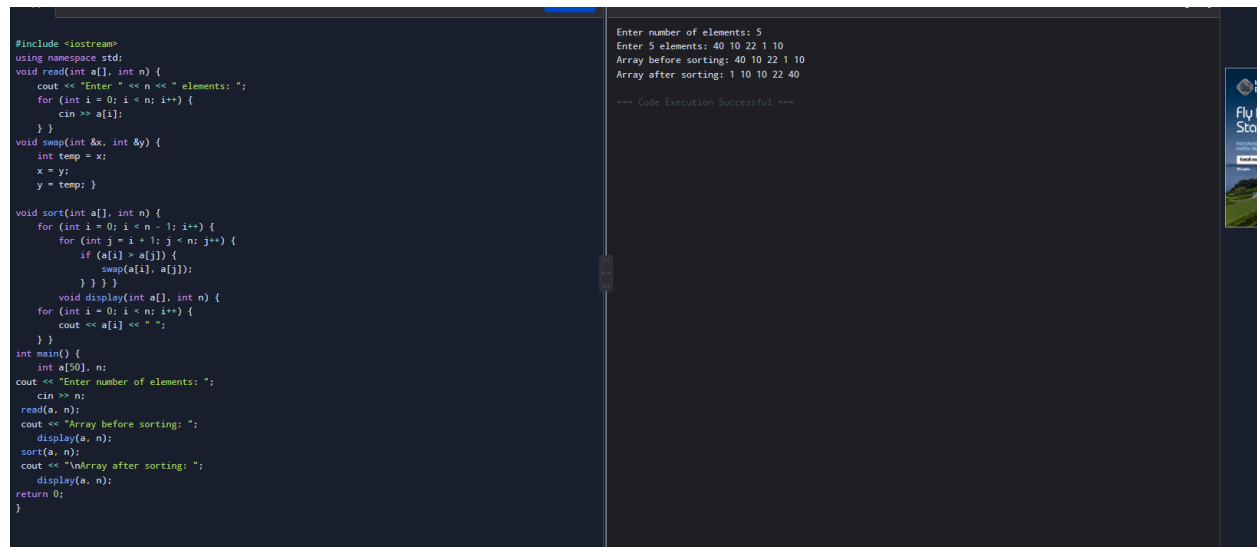
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Program:

```
#include <iostream>
using namespace std;
void read(int a[], int n) {
    cout << "Enter " << n << " elements: ";
    for (int i = 0; i < n; i++) {
        cin >> a[i];
    }
}
void swap(int &x, int &y) {
    int temp = x;
    x = y;
    y = temp;
}
void sort(int a[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if (a[i] > a[j]) {
                swap(a[i], a[j]);
            }
        }
    }
}
void display(int a[], int n) {
    for (int i = 0; i < n; i++) {
        cout << a[i] << " ";
    }
}
int main() {
    int a[50], n;
    cout << "Enter number of elements: ";
    cin >> n;
    read(a, n);
    cout << "Array before sorting: ";
    display(a, n);
```

```
sort(a, n);
cout << "\nArray after sorting: ";
    display(a, n);
return 0;
}
```

Output:



The screenshot shows a C++ program in a code editor on the left and its output in a terminal window on the right. The program implements a bubble sort algorithm. It starts by including `<iostream>` and using the `namespace std;`. It defines three functions: `read` to take input from the user, `swap` to exchange elements, and `sort` to perform the sorting. The `main` function initializes an array of size 50, prompts the user for the number of elements (5), reads the elements (40 10 22 1 10), displays the array before sorting, sorts it, and displays the array after sorting (1 10 10 22 40). The terminal output on the right confirms this, showing the input and the sorted array, followed by a success message.

```
#include <iostream>
using namespace std;
void read(int a[], int n) {
    cout << "Enter " << n << " elements: ";
    for (int i = 0; i < n; i++) {
        cin >> a[i];
    }
}
void swap(int &x, int &y) {
    int temp = x;
    x = y;
    y = temp;
}
void sort(int a[], int n) {
    for (int i = 0; i < n - 1; i++) {
        for (int j = i + 1; j < n; j++) {
            if (a[i] > a[j]) {
                swap(a[i], a[j]);
            }
        }
    }
    void display(int a[], int n) {
        for (int i = 0; i < n; i++) {
            cout << a[i] << " ";
        }
    }
}
int main() {
    int a[50], n;
    cout << "Enter number of elements: ";
    cin >> n;
    read(a, n);
    cout << "Array before sorting: ";
    display(a, n);
    sort(a, n);
    cout << "\nArray after sorting: ";
    display(a, n);
    return 0;
}
```

```
Enter number of elements: 5
Enter 5 elements: 40 10 22 1 10
Array before sorting: 40 10 22 1 10
Array after sorting: 1 10 10 22 40

--- Code Execution Successful ---
```

